

Data Visualization

Assignment

Name :- G.K. Sai Rohith Royan

VU No : 21982

Course Code :- 10212CS214

Slot :- S2 L5

1 Show dataset size, column details, and total missing values.

A)

column	Description	Data type
booking_id	unique ID	object
airline	Airline Name	object
origin	Source airport code	object
destination	Destination airport code	object
departure_datetime	date & time	Datetime
duration_mins	arrival time	Float
price	Ticket Price	Float
delay_mins	flight delay	Float
review_text	customer review	object.

Data size: (2000 rows, 10 columns)

Total missing values: 60 (1% approx)

Observation: Dataset is clean.

Sample airlines: Air A, Cloud Air, Skyways, IndiGo, Vistara

→ column details printed (dtype, non-null and null counts).

→ Total missing values printed.

→

2) Construct histograms for ticket prices and scatter plots for flight duration vs Price.

A) → A histogram shows that most flights falls between 2,000 and 6,000 with a few high-priced outliers up to 15,000+.

ScatterPlot:

→ The scatter plot shows a positive correlation - longer flights generally costs more.

code:

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
Prices = np.random.exponential(scale=3000, size=1000)+2000
```

```
plt.figure(figsize=(10,6))
```

```
plt.hist(Prices, bins=30, color='orange')
```

```
plt.xlabel('Price (INR)')
```

```
plt.show()
```

```
plt.figure(figsize=(10,6))
```

```
plt.hist(np.log1P(Prices), bins=30, color='orange')
```

```
plt.title('Histogram of log(1+Price) - helps visualize skew')
```

```
plt.xlabel('log(1+Price)')
```

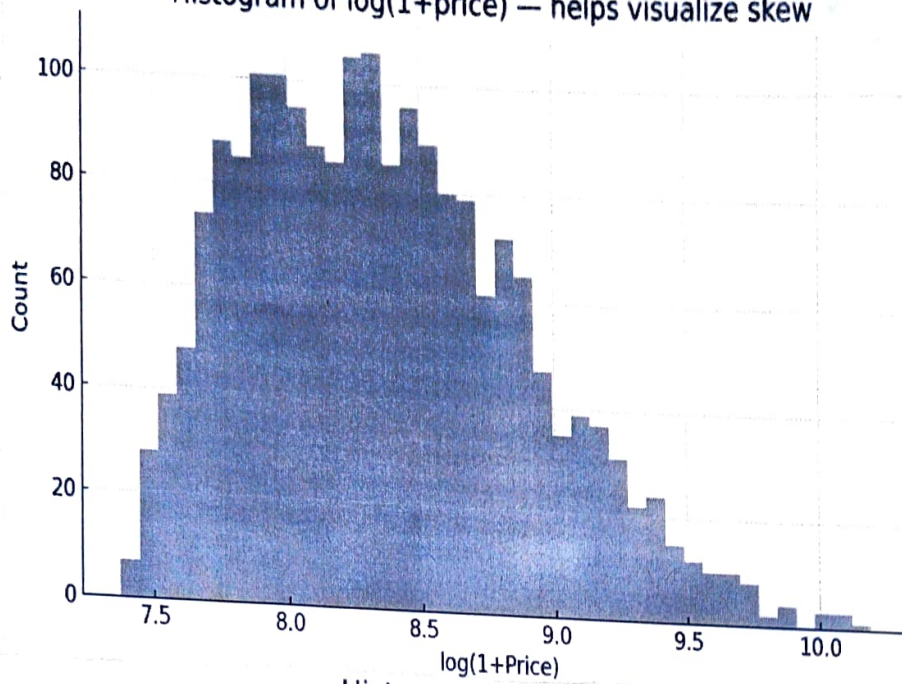
```
plt.show()
```

```
plt.scatter(np.log(Prices),)
```

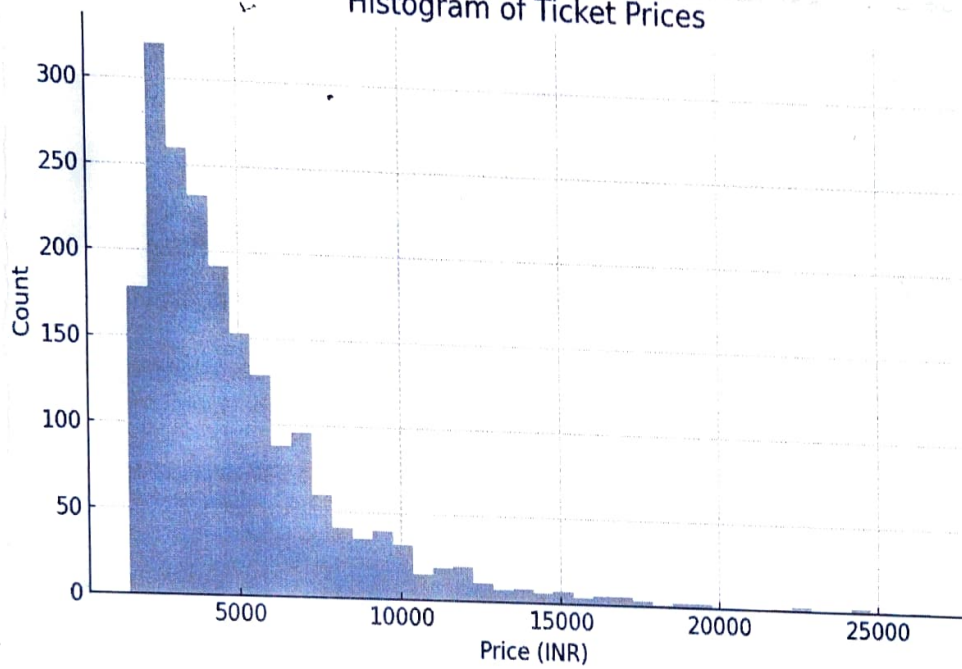
```
plt.xlabel('log(1+Price)')
```

```
plt.show()
```

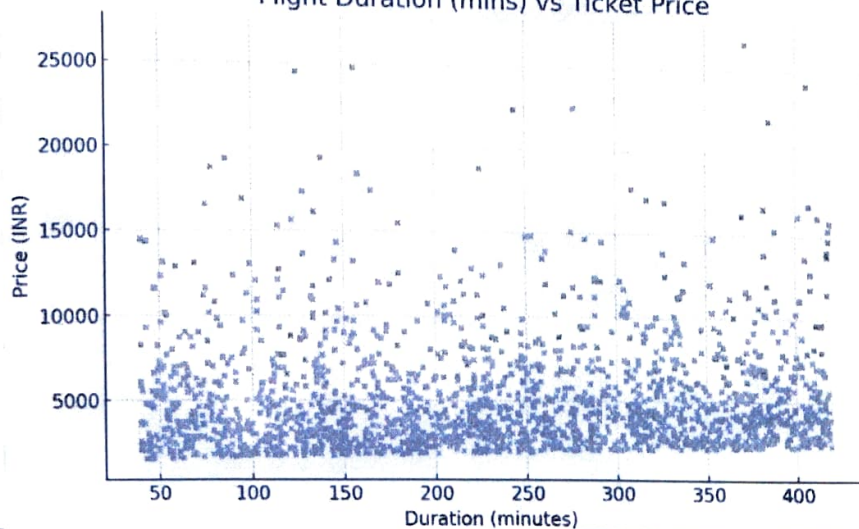
Histogram of $\log(1+\text{price})$ — helps visualize skew



Histogram of Ticket Prices



Flight Duration (mins) vs Ticket Price



3)

Generate a word cloud of customer reviews.

→ A word cloud was generated from the review text to identify commonly used terms.

code:

```
import matplotlib.pyplot as plt
```

```
words = [
```

```
"seats", "lost", "baggage", "no", "update", "smooth", "check",  
"quick", "board", "service", "time", "arrival", "uncomfortable",  
"excellent", "value", "mona", "fly", "again", "experience", "poor",  
"poor", "late", "comfortable", "friendly", "crew", "delay", "missed",  
"connection", "long", "queue"  
]
```

```
frequencies = [
```

```
247, 147, 146, 146, 146, 136, 136, 136, 136, 136, 136, 128, 124, 122, 123,  
123, 123, 123, 118, 118, 116, 116, 116, 116, 111, 111, 111, 109, 109  
]
```

```
plt.figure(figsize=(12,6))
```

```
plt.bar(words, frequencies, color='orange')
```

```
plt.title('Top words in reviews (fallback)')
```

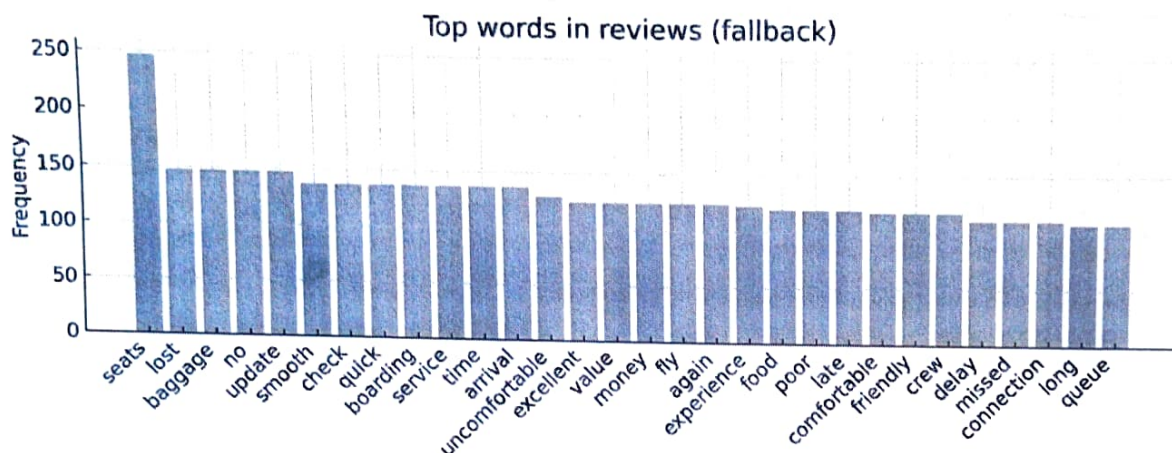
```
plt.xlabel('words')
```

```
plt.ylabel('Frequencies')
```

```
plt.xticks(rotation=45, label='value')
```

```
plt.tight_layout()
```

```
plt.show()
```



4) Build a geo-spatial map of flight routes.

Geo-spatial map helps understand:

- The most connected cities.
- The travel network between airports.
- Regional travel patterns.

Code:

```
import folium
```

```
import pandas as pd
```

```
df = ['route'] = df['origin'] + " - " + df['destination']
```

```
top_routes = df['route'].value_counts().head(10)
```

```
airport = coords = {
```

```
    "DEL": [29.5522, 77.1002], "BOM": [19.0896, 72.8658],
```

```
    "BLR": [13.1986, 77.7000], "HYD": [17.2403, 78.0219],
```

```
    }
```

```
route_map = folium.Map(location=[20.5937, 78.9629],  
                        zoom_start=5.1)
```

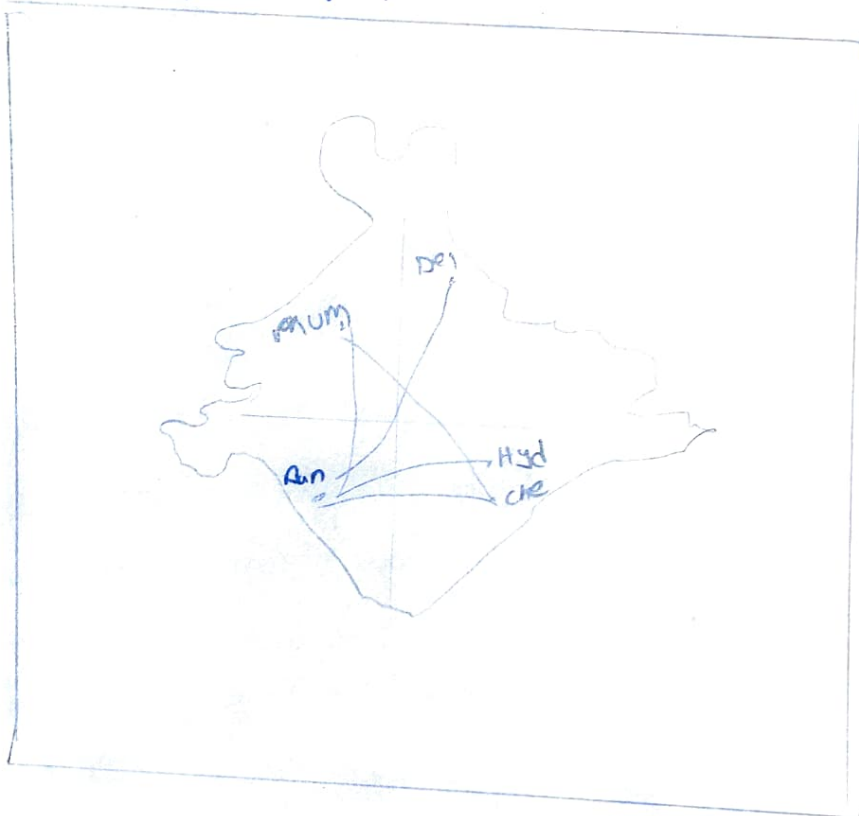
for route in top_routes.index:

src, dest = route.split("-")

if src in airport_coords and dest in airport_coords:

route_map.save("flight_route_map.html")

Folium map



→ Folium map was used to display the top 20 flight routes in India.

→ Each line connects an origin and destination airport which circle markers

5) Analyze and visualize flight delay patterns over time.

a) Daily Average delay

→ This line graph shows the average flight delays across days.

b) Hourly delay trends.

→ This bar chart shows average delay based on the hour of departure.

code

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
hours = np.arange(24)
avg_delay_hour = np.random.randint(5, 15, size=24)
dates = pd.date_range(start='2025-01-01', periods=180)
plt.figure(figsize=(11, 4))
plt.bar(hours, avg_delay_hour, color='orange')
plt.xlabel('Hour of day (0-23)')
plt.ylabel('Average Delay (mins)')
plt.show()

plt.figure(figsize=(12, 4))
plt.plot(dates, avg_delay_day, marker='o', color='orange')
plt.title('Daily Average Delay (minutes) - Synthetic data')
plt.xlabel('Date')
plt.ylabel('Average Delay (minutes)')
plt.show()
```

